



LESSON - 5



SIKHOGE KYA???

INTRODUCTION & KEY POINTS

- INTRODUCTION
- OVERVIEW OF DATABASES
- APPLICATIONS & JOBS
- SOFTWARE INSTALLATION -
- MYSQL SERVER AND MYSQL WORK BENCH WITH DATABASE SETUP
- TABLES CREATION & FIRST QUERY

SQL STATEMENT FUNDAMENTALS

- SELECT * FROM
- SELECT DISTINCT
- COUNT, ALIAS AS
- SELECT WHERE
- LIMIT AND ORDER BY
- OPERATORS AND FILTERING
- BETWEEN, IN, LIKE AND ILIKE
- CASE IF ELSE
- COALESCE, NULLIF



JOINS & SUB QUERY

- INNER JOIN
- LEFT JOIN (LEFT OUTER JOIN)
- RIGHT JOIN (RIGHT OUTER JOIN)
- FULL JOIN (FULL OUTER JOIN)
- CROSS JOIN
- SELF JOIN
- -----
- STRING FUNCTIONS AND OPERATORS
- SUBQUERY - IMPORTANT
- DATE & TIME FUNCTIONS

GROUP BY & HAVING

- INTRODUCTION TO GROUP BY
- AGGREGATION FUNCTIONS
- GROUP BY - CHALLENGE
- HAVING
- HAVING - CHALLENGE TASKS

DATA MANIPULATION (DML)

- INSERT
- UPDATE
- DELETE
- COPY RENAME STRUCTURE
- CHECK CONSTRAINT

DATA DEFINITION LANGUAGE (DDL)

- DATA TYPES & ROWS COLUMNS
- PRIMARY FOREIGN & OTHER KEYS
- CREATE TABLE
- ALTER TABLE
- DROP TABLE
- CONSTRAINTS

ADVANCE SQL

- VIEWS / TEMP TABLES & INDEXES
- FUNCTIONS & TRIGGERS
- STORE PROCEDURES
- TRANSACTIONS
- WINDOW FUNCTIONS
- QUERY ANALYSIS TECHNIQUES

Keys in SQL

Keys are one of the basic requirements of a relational database model. It is widely used to identify the tuples(rows) uniquely in the table.

Primary Key

A unique identifier for each record in a table.
Ensures that no duplicate values or NULL values are present.

A table can have only one primary key,
which can consist of a single column or a combination of columns (composite key).

```
CREATE TABLE Customers (
    CustomerID INT PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100)
);
```

Foreign Key

Used to link two tables together.
Refers to the primary key in another table, establishing a relationship.

```
CREATE TABLE Orders (
    OrderId INT PRIMARY KEY,
    ProductID VARCHAR(50),
    OrderDate Date,
    CustomerID INT,
    FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);
```

Unique Key

Ensures all values in a column are unique but allows one NULL value.
A table can have multiple unique keys.

```
CREATE TABLE Users (
    UserId INT PRIMARY KEY,
    Name VARCHAR(50),
    Email VARCHAR(50) UNIQUE
);
```

Candidate Key

A column or set of columns that can qualify as a unique identifier.
A table can have multiple candidate keys, but only one can be chosen as the primary key.

Example
In a table with columns StudentID and Email, both could potentially be candidate keys.

Alternate Key

A candidate key that is not chosen as the primary key.
Still unique and can be used to access records..

PRACTICE / ASSIGMENT

Here are some practical DDL questions to help students understand and apply DDL concepts.

Create a table Students with the following columns:

StudentID - PRIMARY KEY
FirstName (String, 50 characters, Not Null)
LastName (String, 50 characters, Not Null)
DateOfBirth (Date)
ClassID

Create a table Classes with the following structure:

ClassID (Integer, Primary Key)
ClassName (String, 100 characters, Not Null)
TeacherID (Integer, Foreign Key referencing TeacherID in the Teachers table)

PRACTICE / ASSIGMENT - khtm by by tata gya

Here are some practical DDL questions to help students understand and apply DDL concepts.

2. ALTER Statements

1. Add a column Email (String, 100 characters) to the Students table.
2. Modify the ClassName column in the Classes table to allow 200 characters.
3. Add a foreign key ClassID in the Students table referencing ClassID in the Classes table.
4. Drop the Subject column from the Teachers table.

3. DROP Statements

1. Drop the Teachers table from the database.
2. Delete the index created on the LastName column in the Students table.

4. TRUNCATE Statements

1. Write a query to remove all rows from the Students table while keeping its structure intact.